

*MA.7.GR.1.4***Benchmark****MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.**

Example: If a 12-inch pizza is cut into 6 equal slices and Mikel ate 2 slices, how many square inches of pizza did he eat?

Benchmark Clarifications:

Clarification 1: Instruction focuses on the connection between formulas for the area of a rectangle and the area of a circle.

Clarification 2: Problem types include finding areas of fractional parts of a circle.

Clarification 3: Solutions may be represented in terms of pi (π) or approximately.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.1
- MA.7.AR.4.1
- MA.7.GR.2.1, MA.7.GR.2.2, MA.7.GR.2.3

Terms from the K-12 Glossary

- Area
- Circle
- Pi (π)
- Rectangle

Vertical Alignment**Previous Benchmarks**

- This is the first introduction to circles outside of recognizing the shape in early elementary.

Next Benchmarks

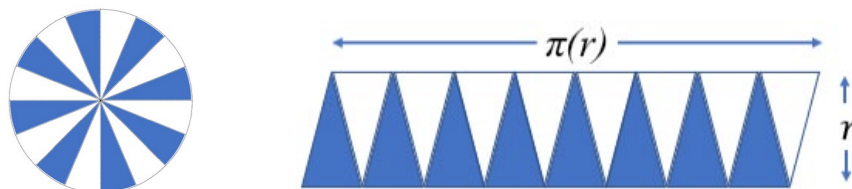
- MA.912.GR.4.4, MA.912.GR.4.6
- MA.912.GR.6.4

Purpose and Instructional Strategies

In grade 6, students found the areas of rectangles and triangles, and solved problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles. In grade 7, students find the area of circles and other geometric figures. In both grade 7 and high school, students build on their knowledge of area to find the surface areas and volumes of various three-dimensional figures.

- Students are not expected to memorize the formula for the area of a circle (*MTR.5.1*).
- Students are expected to know approximations of pi ($\frac{355}{113}$, $\frac{22}{7}$ or 3.14).
- Instruction includes students exploring circles. Provide students with a circle and have them highlight the circumference. Students will then fold the circle in half, half again, and half once more to allow them to cut it into 8 wedges of equal size. Then arrange the wedges so they alternately point up and down, forming a rectangle, with the highlighted circumference being the bases. Ensure students realize that the length of the rectangle is approximately equal to half the circumference, or πr , of the circle and the height of the rectangle is equal to the radius, r , of the circle (*MTR.4.1*).

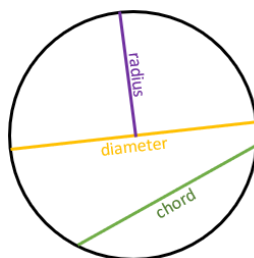
- For more accuracy, provide a circle with lines drawn for students to cut the circle into 16 equal-sized wedges.



- Have students describe the area of a circle and explain if the area of a circle changes if it is cut up and rearranged.
- Ask questions to elicit student thinking (*MTR.4.1*) such as:
 - What formula was used to find the area of a circle?
 - How is the formula for the area of a circle related to the formula for the area of a parallelogram?
- Instruction includes using circles on grid paper for students to estimate area before making precise calculations.
- The expectation of this benchmark is not to find the radius or diameter of a circle when given the area.

Common Misconceptions or Errors

- Students may invert the terms radius and diameter. To address this misconception, review parts of a circle including radii, diameters and chords.

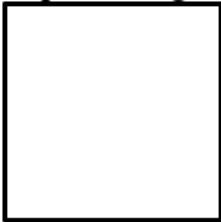


- Students may incorrectly believe pi is a variable, rather than a constant that does not change from one circle to the next. Review the development of pi in MA.7.GR.1.3.
- Students may confuse circumference and area. To address this misconception, help students connect area of a circle to area of a rectangle.
- Students may incorrectly double the radius (finding diameter), rather than squaring it, when finding an area. To address this misconception, review exponent rules from MA.6.NSO.3.3 and MA.7.NSO.1.1.

Strategies to Support Tiered Instruction

- Teacher provides opportunities for students to measure the radius and diameter of various circles and to explore and discuss the similarities and differences between radius and diameter.
- Instruction includes modeling the area of a given circle in terms of pi before replacing the value of pi with an approximation to determine the estimated area.

- Teacher co-creates a graphic organizer with students containing color-coded examples of circumference, area, diameter and radius.
- Teacher co-creates a table to find the constant of proportionality between the diameter and circumference of a circle, allowing students to discover the pattern that represents pi (π).
- Instruction includes modeling the differences between doubling and squaring a radius. Doubling a radius would be represented by multiplying the given length by 2, whereas squaring a number would be represented by the area of a square with the given radius.
 - For example, students can be given the table below to show how the left column doubles a length whereas the right column squares a length.

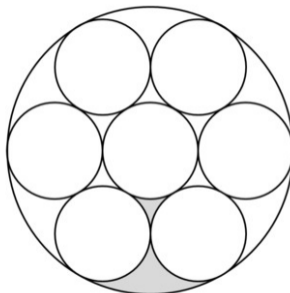
Doubling	Squaring
Given length of 5 _____	Given length of 5 _____
Representing $5 \cdot 2$ _____	Representing 5^2 

- Instruction includes rewriting the area formula for a circle in expanded form before evaluating.
 - For example, the formula for the area of a circle, $A = \pi r^2$, can be rewritten as $A = (\pi)(r)(r)$.
- Teacher helps students connect area of a circle to area of a rectangle.

Instructional Tasks

Instructional Task 1 (MTR.1.1)

The figure below is composed of eight circles, with seven small circles inside one large circle containing them all. Neighboring circles only share one point, and two regions between the smaller circles have been shaded. Each small circle has a radius of 5 centimeters.



Part A. What is the area of the large circle?

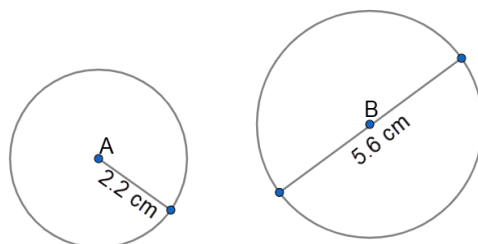
Part B. What is the area of the shaded part of the figure?

Instructional Items*Instructional Item 1*

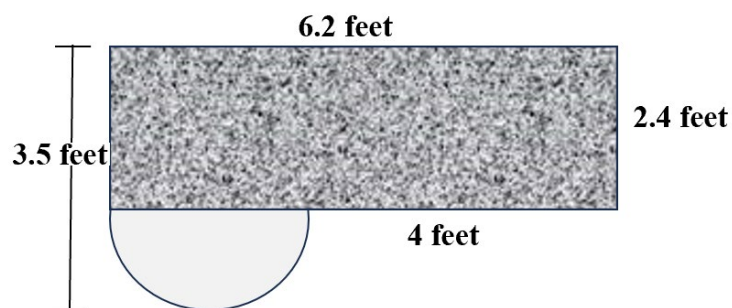
What is the area of a circle whose radius is 4 centimeters? Round to the nearest hundredth.

Instructional Item 2

Find the exact area, in centimeters (cm), of each circle below.

*Instructional Item 3*

Jamilah wants to add to her kitchen countertop, which is currently in the shape of a rectangle. If she adds the solid, semicircular piece shown in the picture below, determine how many square feet, to the nearest tenth of a foot, of marble Jamilah will need for the addition.



**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*